Project Initialization and Planning Phase

Date	2 July 2026
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to optimize agricultural yield and resource efficiency by implementing advanced machine learning techniques to recommend the most compatible crops based on soil and weather parameters.
Scope	The project comprehensively models 22 different crop classes against Nitrogen, Phosphorous, Potassium, temperature, humidity, soil pH, and rainfall to provide instant, real-time farm recommendations.
Problem Statement	
Description	Address the guesswork and inefficiencies in traditional farming, which lead to soil degradation, resource waste, and crop failure.
Impact	Improves crop yield, prevents over-fertilization, increases farming profits, and provides policymakers with data-driven agricultural planning insights.
Proposed Solution	
Approach	Employing a Random Forest Classifier trained on 2,200 observations to predict the optimal crop, combined with statistical checks to gauge soil-climate compatibility.
Key Features	• ML Crop Recommendation Model (99.32% accuracy) • Compatibility assessment engine with alignment gauges • Interactive agricultural research dashboard

	Real-time decision-making for optimal crop selection. Data-driven advice to adapt to changing seasonal weather envelopes.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Modern Multi-core CPU
Memory	RAM specifications	8 GB RAM
Storage	Disk space for data, models, and logs	500 MB Free Disk Space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Agricultural crop dataset, 2200 rows, CSV format